OTTO VON GUERICKE UNIVERSITÄT MAGDEBURG

INF FAKULTÄT FÜR INFORMATIK

Fraunhofer IFF

Master's Thesis Proposal

Pre-trained Deep Learning Networks for Hyperspectral Image Data

(Developed in collaboration with Fraunhofer Institute for Factory Operation and Automation (IFF)**)**

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Background and Context

Industry Partnership: Addressing real-world HSI data challenges at **Fraunhofer IFF**.

The Data: HSI cubes with up to **256 spectral channels**—capturing precise material "fingerprints" but increasing complexity.

The Problem: Current models must be trained **from scratch** for every new task, which is slow and inefficient.

The Gap: Unlike RGB vision (which uses pre-trained models like **ViT or CLIP**), HSI lacks a universal "backbone" to jumpstart new tasks.

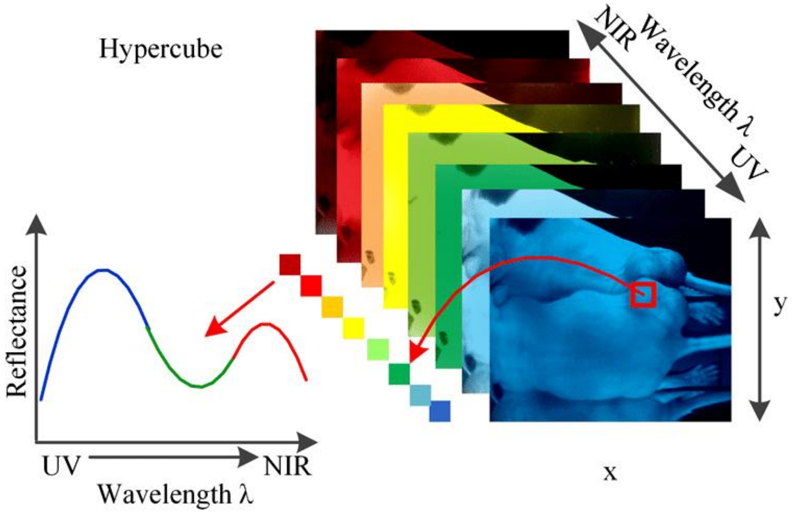


Fig Source: Vishvanathan, Sowmya & Kp, Soman & Hassaballah, Mahmoud. (2019).

Research Objectives

RQ1: How does a discriminative objective (SimCLR-based Contrastive Learning) a strategy not explored in the baseline paper (Scheibenreif et al.) compare to the generative approach (MAE) used in previous HSI research?

RQ2: Which pre-training strategy provides the most significant reduction in labeled data requirements while maintaining high classification accuracy for real-world tasks?

RQ3: How does our self-supervised pretraining compare to state-of-the-art hyperspectral foundation models on downstream classification?

Data and Processing

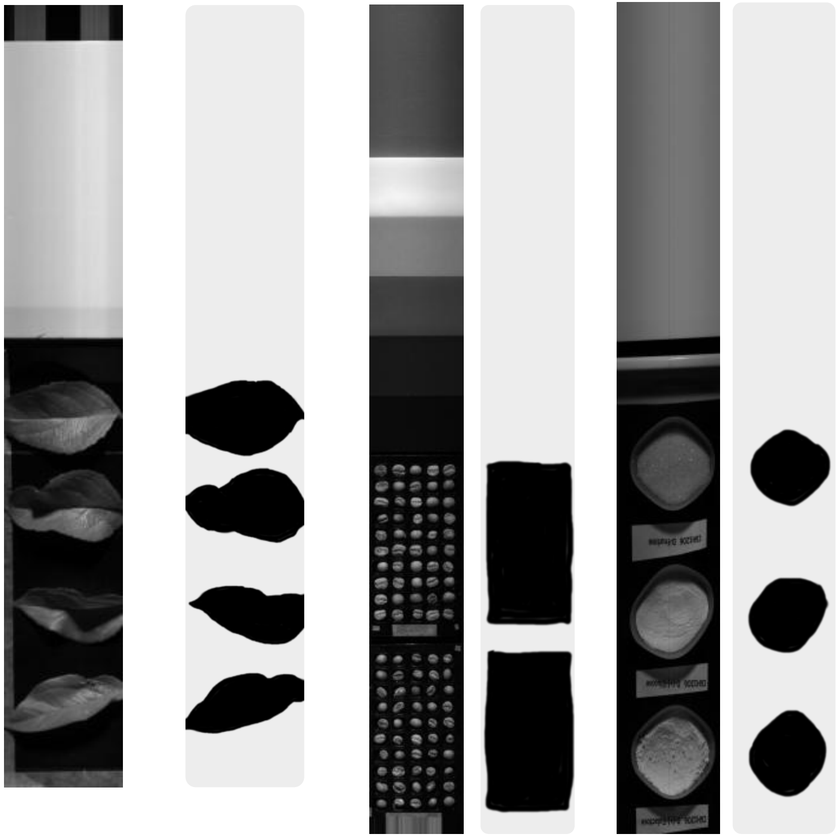
Real-world datasets provided by Fraunhofer IFF

Sourced from multiple domains (coffee, leaf, grapes, sugar, paper, archeology, branches),
totaling ~5,000 spectrally-rich samples.
Processing Steps:

* Image Calibration
* Cropping and background masking
* Patch Creation

VNIR (Visible & Near-Infrared) and SWIR (Short-Wave Infrared) sensors
SWIR - 256,288
VNIR - 160

Sample Patches– 64x64x256 (interpolate)



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Thesis Task: Implementation & Validation

Find suitable architecture: Select and adapt Vision Transformer for HSI domain

Process data: Prepare institutional HSI datasets for pre-training (patching, calibration)

Implement two methods:
Method 1: Masked Image Modelling (Scheibenreif et al., 2023)
Method 2: Contrastive Learning (Hu et al., 2021)

Train & validate: Pre-train both networks and fine-tune on downstream tasks.

Vision Transformer

The Vision Transformer (ViT), introduced by Dosovitskiy et al. (2021), shifted computer vision from local convolutions to global attention. Unlike CNNs that use repetitive filters, ViTs treat images as sequences of patches, allowing for superior relationship modeling.

Key Features:
Patch Embedding: Images are divided into fixed-size patches and processed as linear sequences.
Global Attention: Every region interacts with every other region simultaneously, regardless of distance.
Long-Range Dependencies: Captures complex spatial-spectral relationships that CNNs often miss due to their limited receptive field.
Transformer Scaling: Provides a flexible architecture that scales efficiently with large, high-dimensional datasets.

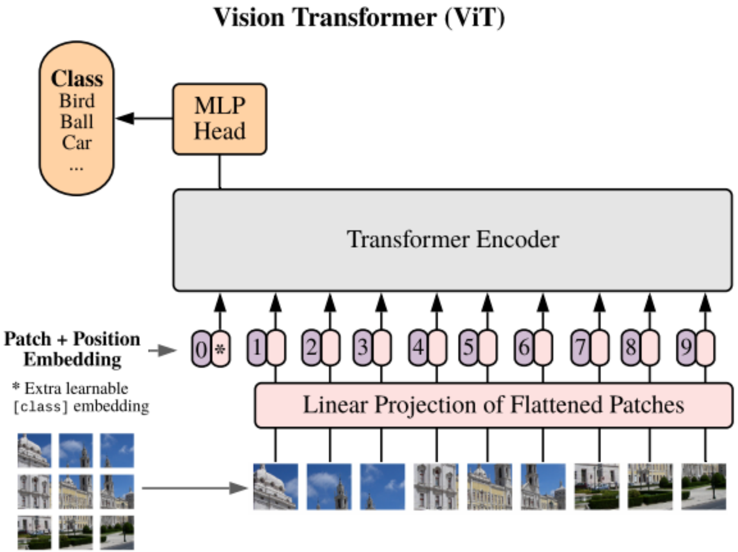
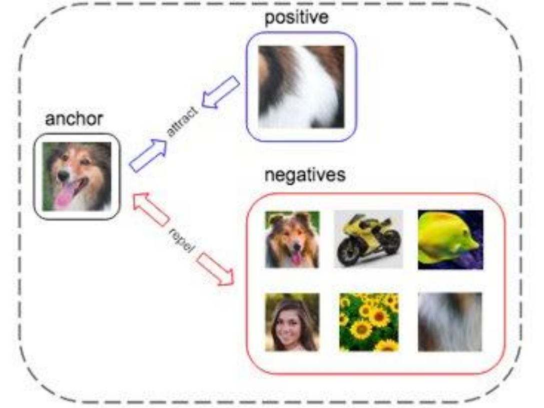


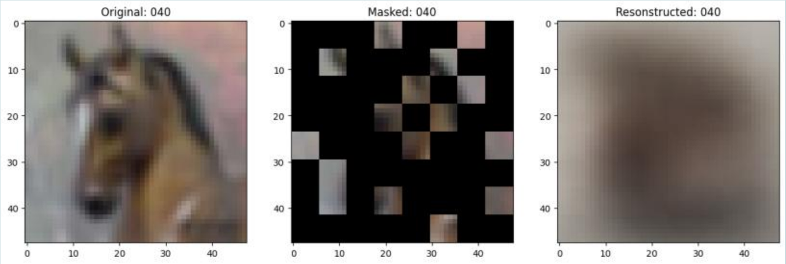
Fig Source: Adapted from Dosovitskiy et al. (2021)

Contrastive Learning



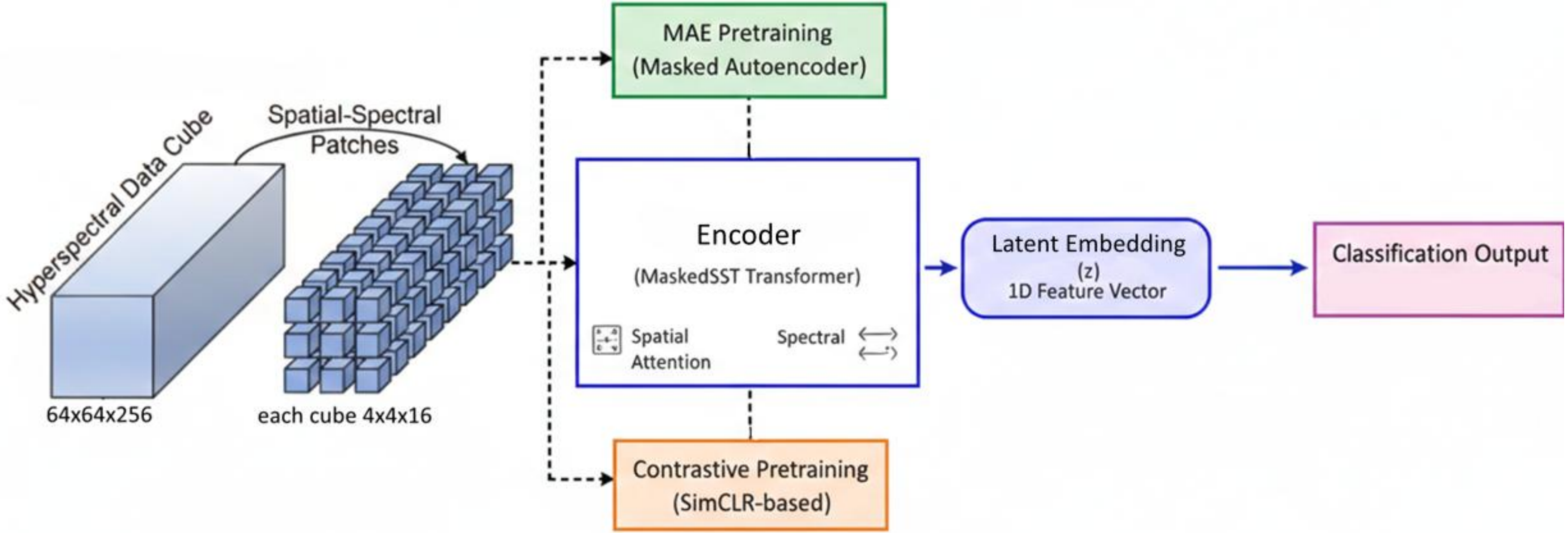
This is a **discriminative** approach where the model learns by comparing different "views" of the same data. Two different augmentations (like rotations or color shifts) are created from a single HSI patch, and the model is trained to recognize that these views represent the same object while pushing them away from "negative" samples. This forces the network to extract high-level, invariant features that are crucial for distinguishing between different classes.

Masked Image Modelling

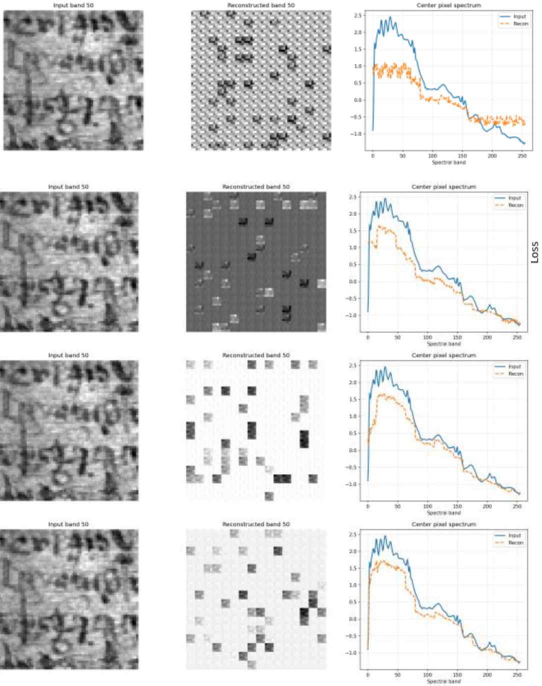


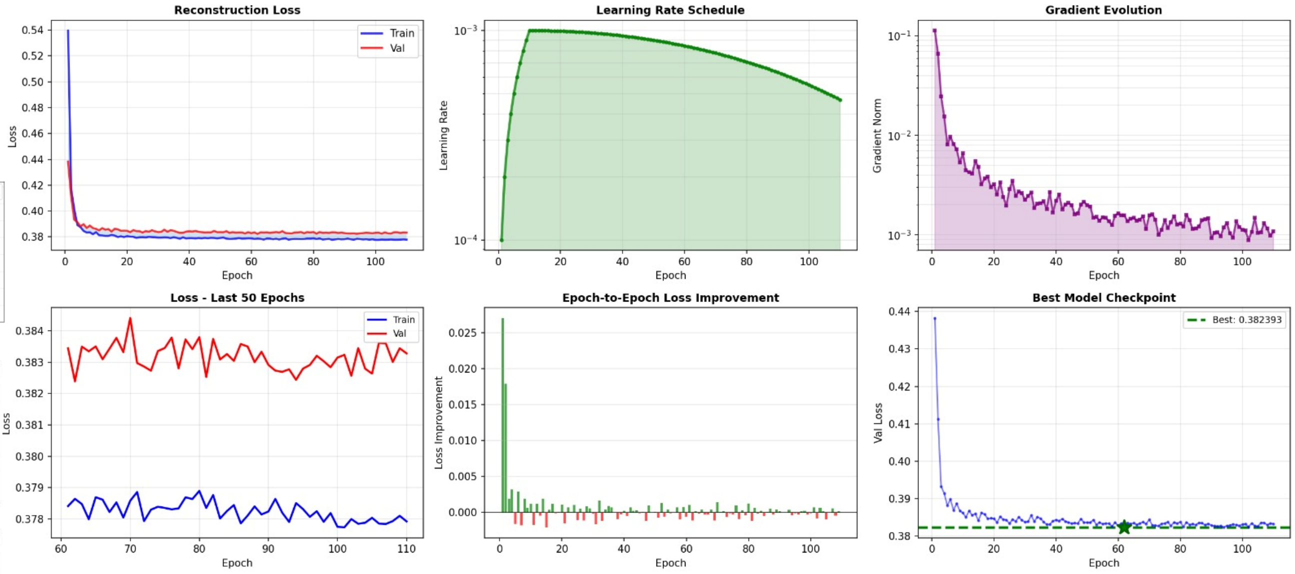
This is a **generative** approach where the model learns by reconstructing "hidden" parts of an image. A large portion of the input patches are masked out, and the transformer is forced to predict the missing pixels or spectral values by understanding the context and relationships within the visible patches. This teaches the network the fundamental spatial-spectral structure of the data.

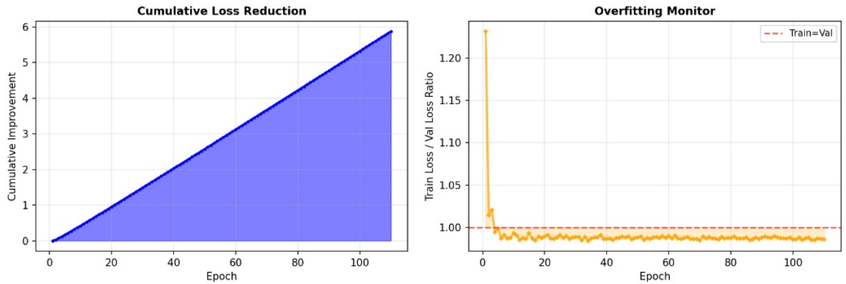
Unified Learning Pipeline



MAE - Masked Autoencoder





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TRAINING SUMMARY
Total Epochs: 110

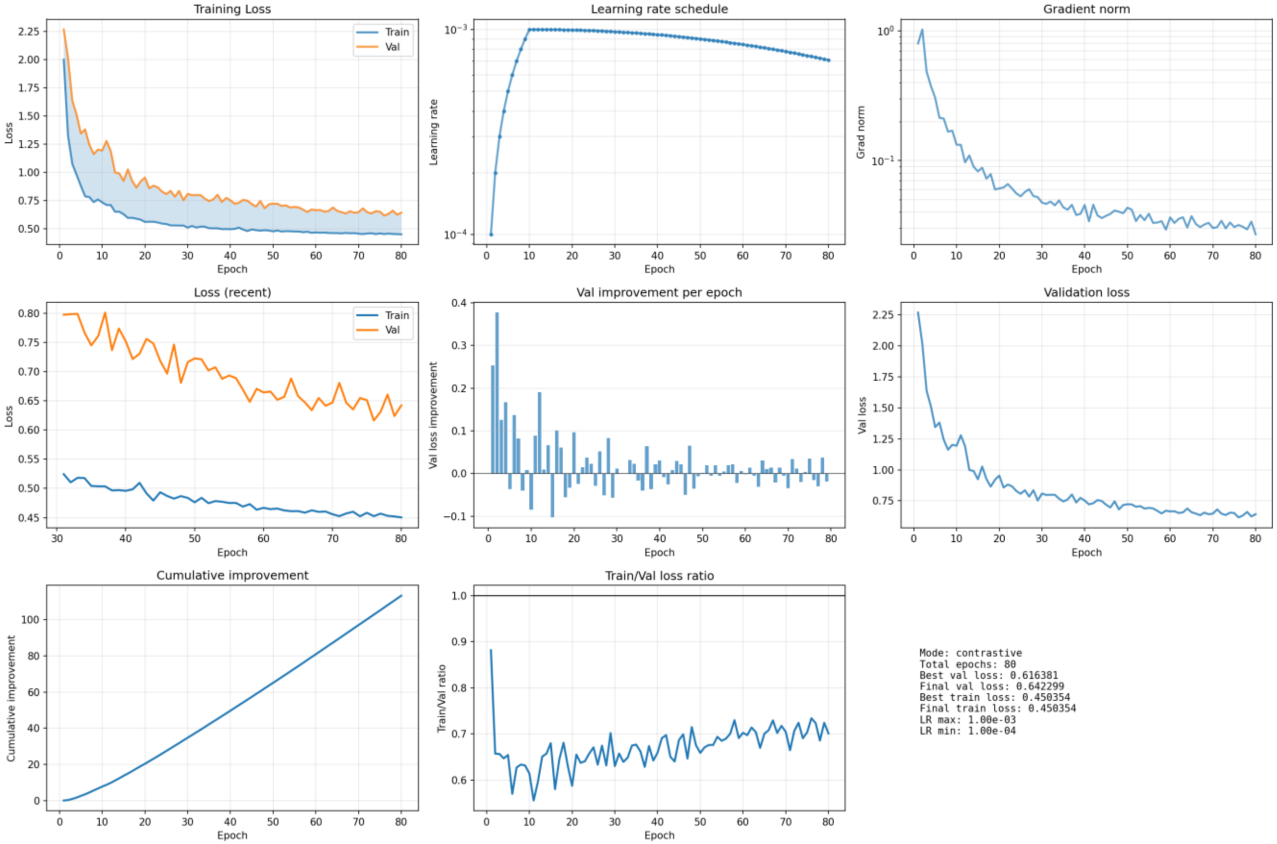
Best Val Loss: 0.382393
Final Val Loss: 0.383285

Best Train Loss: 0.377731
Final Train Loss: 0.377918

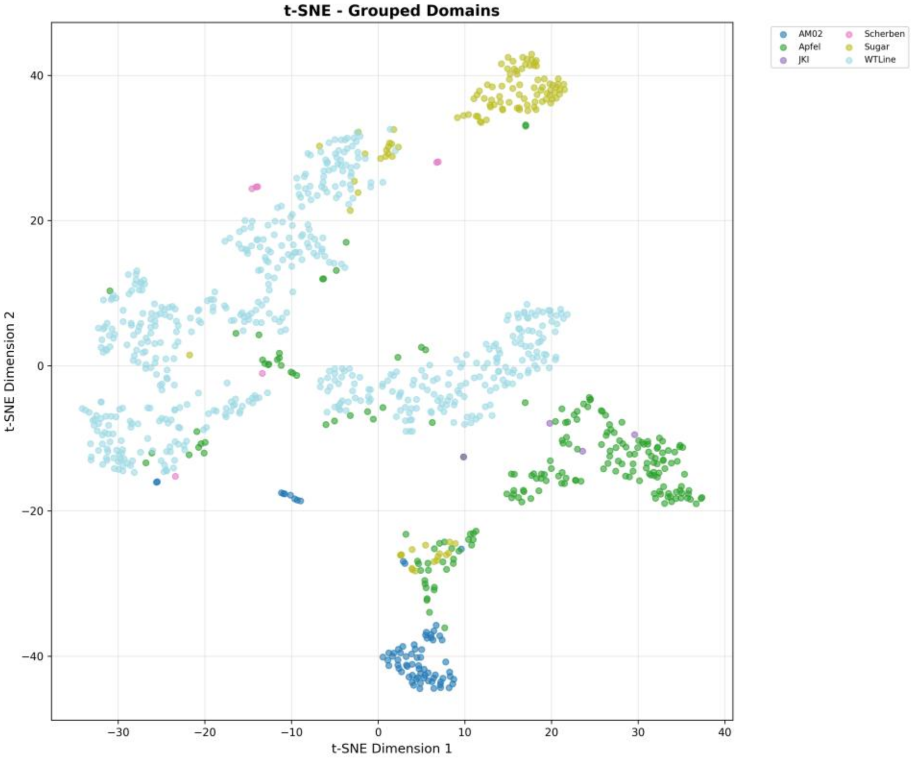
Loss Reduction: 0.055857
(12.7%)

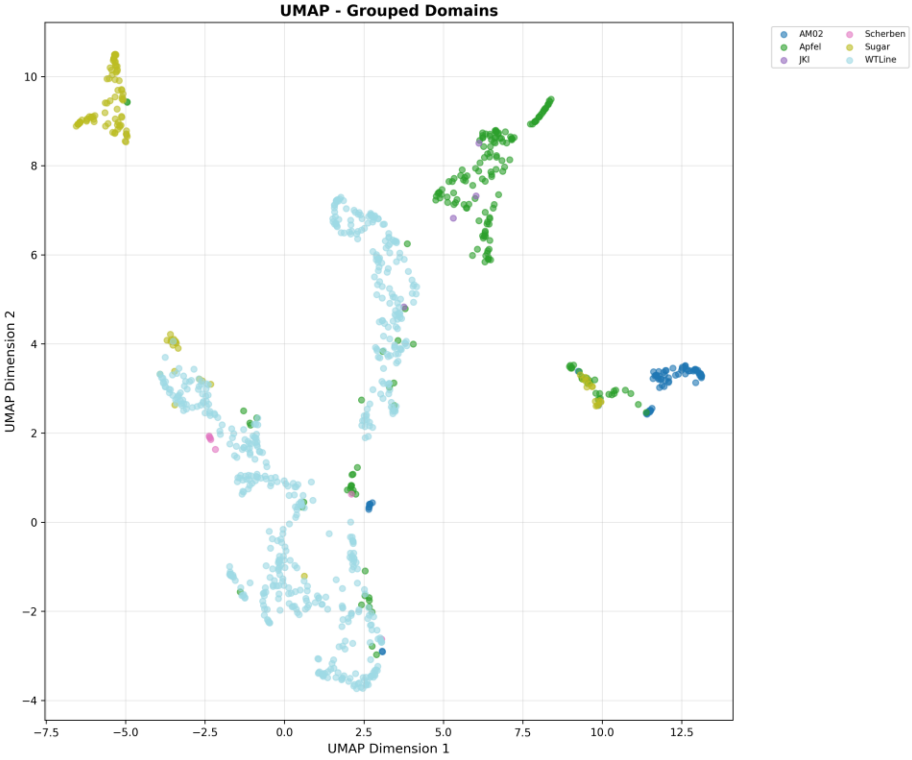
Learning Rate Range:
Max: 1.00e-03
Min: 1.00e-04
```

Contrastive learning



T-SNE and UMAP visualization for mae





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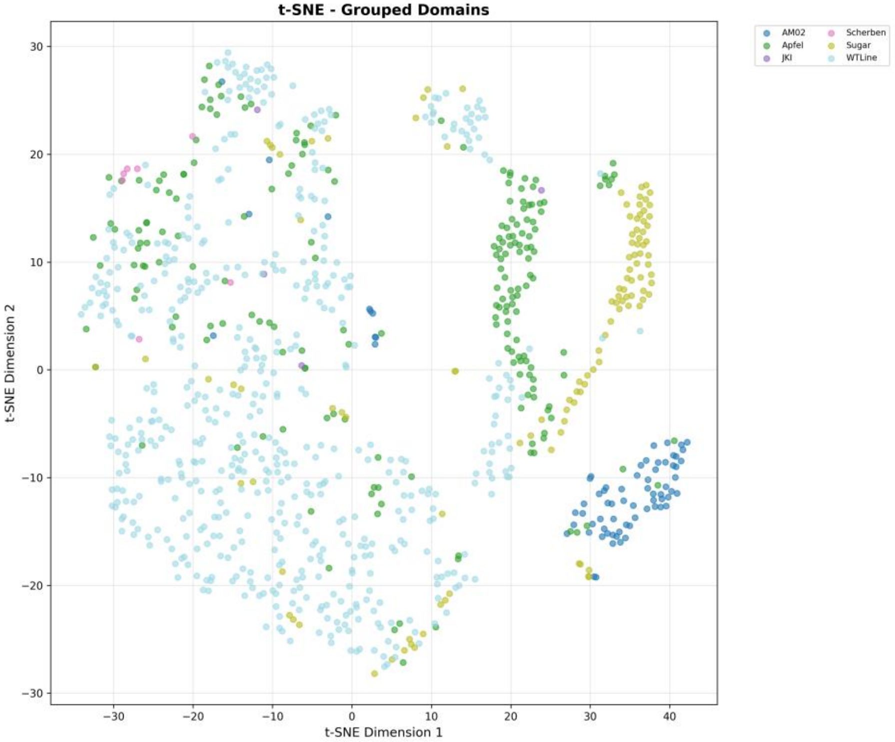
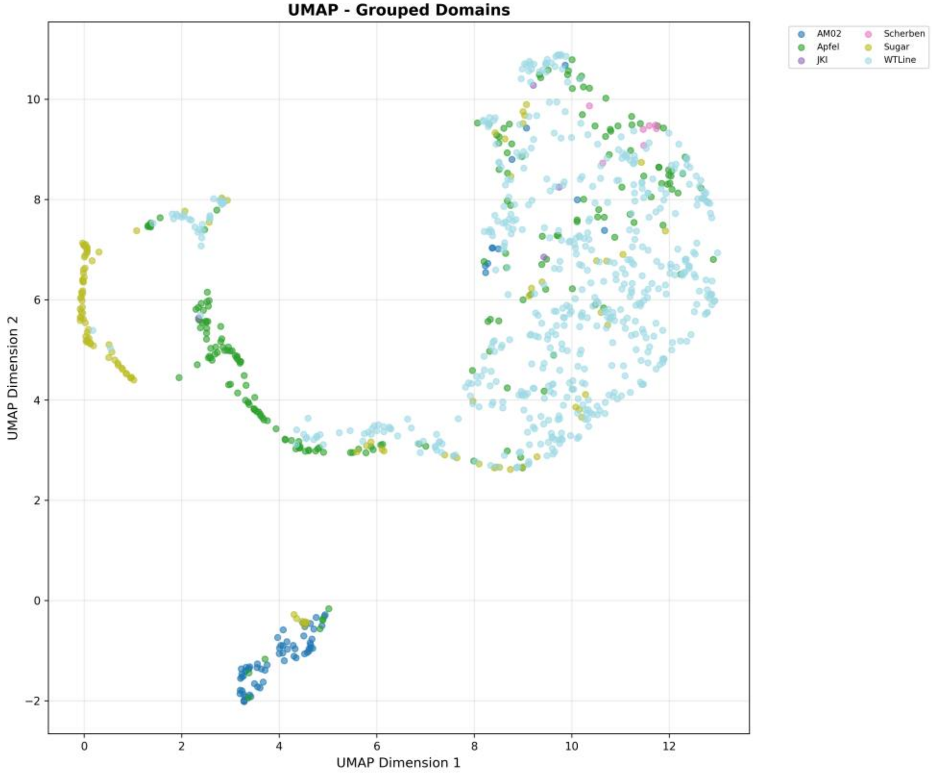
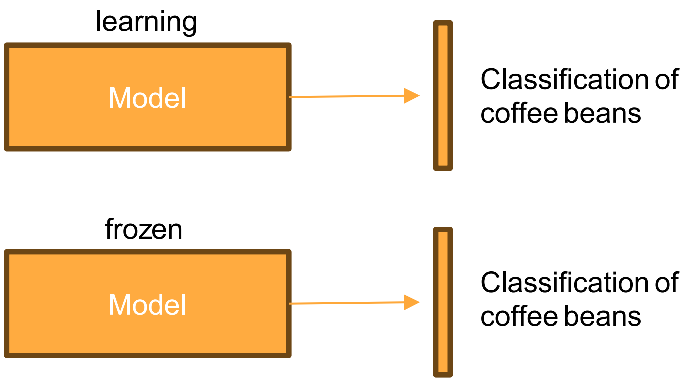
T-SNE and UMAP visualization for contrastive





Random Initialization

MAE/Contrastive



Pre-training Method	Validation Accuracy	Improvement over Random Init	Relative Gain
Random Initialization	**31.13%**	Baseline	-
MAE (Reconstruction)	**52.17%**	**+21.04%**	**+68%**
Contrastive (SimCLR)	**60.87%**	**+29.74%**	**+96%**

Masked Vision Transformers for Hyperspectral Image Classification(https://github.com/HSG-AIML/MaskedSST)

On EnMAP Dataset:

- **Without pretraining:** 74% accuracy
- **With masked pretraining:** 79% accuracy (+5%)

On Houston2018 Dataset:

- **Without pretraining:** 33% accuracy
- **With masked pretraining:** 48% accuracy (+15%)

Method	Pretraining Strategy	Pretraining Data	Apple Test Acc.	Reference
Random Init	None	-	?	Baseline
Ours (MAE)	Self-supervised MAE	Coffee + Grapes + Pottery	**?**	**Our work**
Ours (SimCLR)	Self-supervised Contrastive	Coffee + Grapes + Pottery	**?**	**Our work**
MaskedSST (Houston)	Masked self-supervised	Houston2018 (urban HSI)	?	[Scheibenreif et al., 2023]
MaskedSST (EnMAP)	Masked self-supervised	EnMAP (satellite HSI)	?	[Scheibenreif et al., 2023]

Class	Description	PCR Result	Visual Symptoms
1	**Infected** (positiv)	Positive PCR	Visible red coloration symptoms on leaves
2	**Infected** (negativ)	Negative PCR	Red/light-red symptoms but PCR negative
3	**Healthy** (negativ)	Negative PCR	Green leaves, no symptoms
4	**Infected-asymptomatic** (positiv)	Positive PCR	Green leaves but PCR positive

Trees: 111+ individual trees sample
Leaves per tree: 4 leaf replicates per tree

Total patches: 5881

Model	Download Link	Pretrained Data	Expected Apple Accuracy	Code Availability
SpectralGPT	GitHub Releases	Large-scale RS	**?**	Full code provided
SpectralFormer	GitHub	HSI benchmarks	**?**	Full code provided

Training Infrastructure

Hardware Setup
Training is performed on the university cluster with 8× Tesla V100 GPUs (32GB each)
Multi-GPU Training:
Using DataParallel, we distribute the batch across all 8 GPUs. Each GPU processes 8 samples independently, then gradients are synchronized.

Training Speed:
- Mae - 8 GPUs: ~90 -120 min for 100 epochs
- Cont - 8 GPUs: ~200min for 100 epochs

Batch Size & Memory:
Total batch: 64 samples
Per GPU: 8 samples

Reference

- **Baseline Paper:** Scheibenreif, L., et al. (2023). *Masked Vision Transformers for Hyperspectral Image Classification.*
- **ViT Architecture:** Dosovitskiy, A., et al. (2021). *An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale.*
- **SimCLR Method:** Chen, T., et al. (2020). *A Simple Framework for Contrastive Learning of Visual Representations.*

Thank You!